

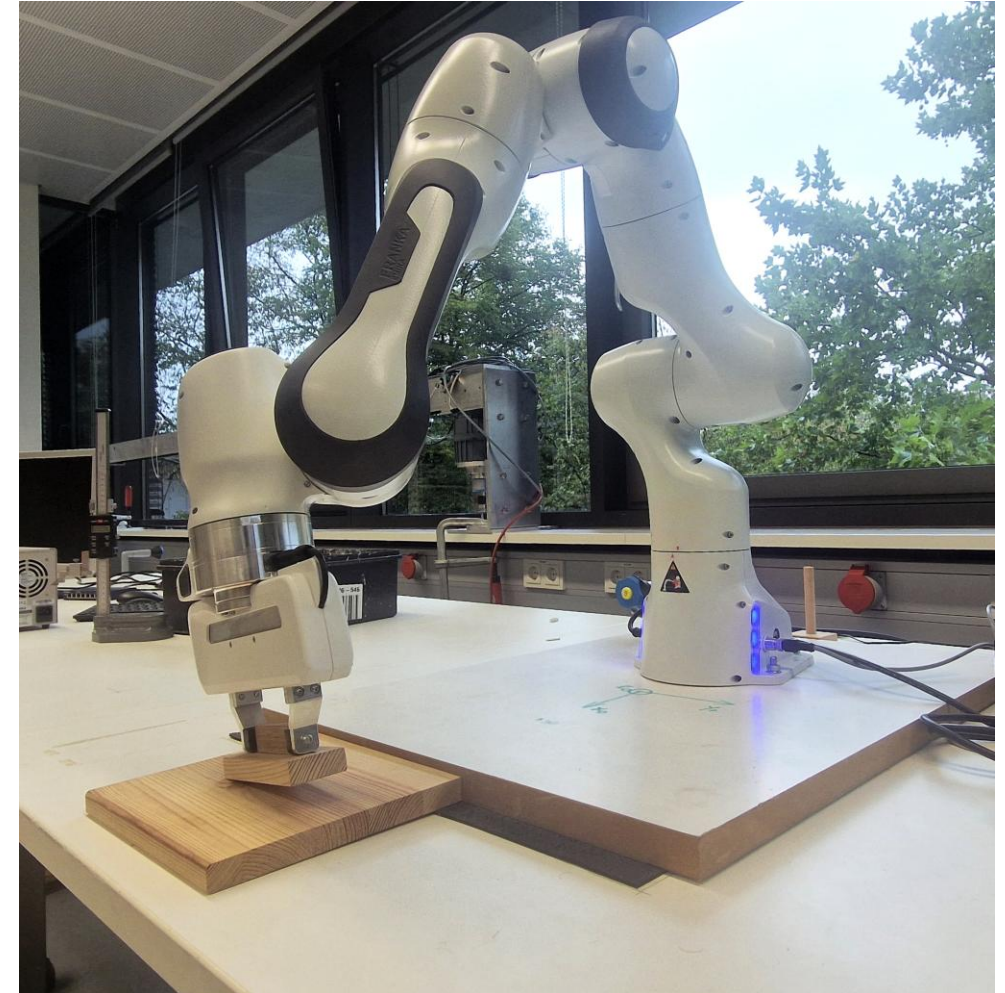
Design and Implementation of a Real-Time Impedance Controller for a 7-DOF Collaborative Robot

Heykel Khadhraoui

Computational Engineering (M.Sc.)

Supervisors: Prof. Norbert Nitzsche · Prof. Ruth Otto

Date: 05.10.2026



-
- $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$
 $\rightarrow P, R$ state. q
 $q, \dot{q} \rightarrow$ state. dq
 $(p, r):$ $CP = P_d - P_{ss}$
 $\Delta R = R_{EE}^T R_{dd}$
 $R_{R,EE} = \frac{1}{U} \rightarrow$ Drehaxe
 $\rightarrow$ Drehwinkel
 $KS \quad CR = R_{EE}^T R_{R,EE}$

Overview

1. **Theory**
2. **Cartesian controller**
3. **Contact experiments**
4. **Contact results**
5. **Null-space control and experiments**
6. **Conclusion and future work**

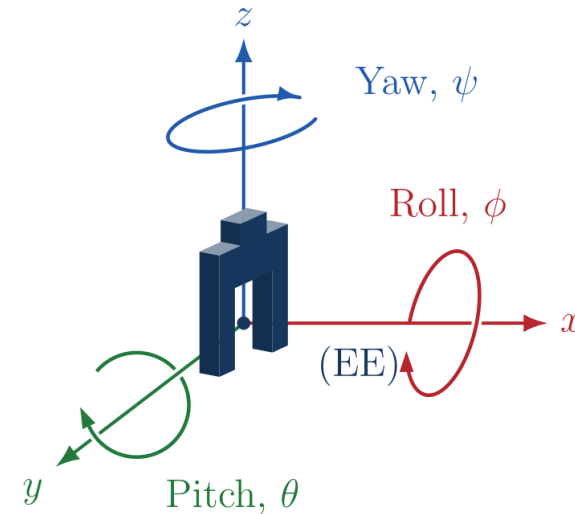
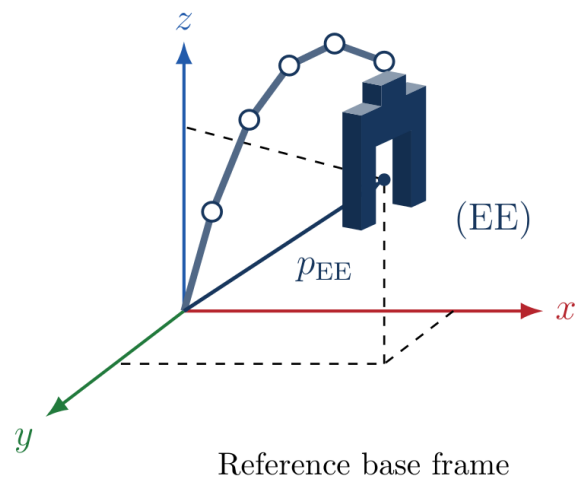
Cartesian pose

Cartesian pose of the end-effector (EE):

$$x_{EE} = [x \quad y \quad z \quad \varphi \quad \theta \quad \psi]^T$$

Position: 3 translations

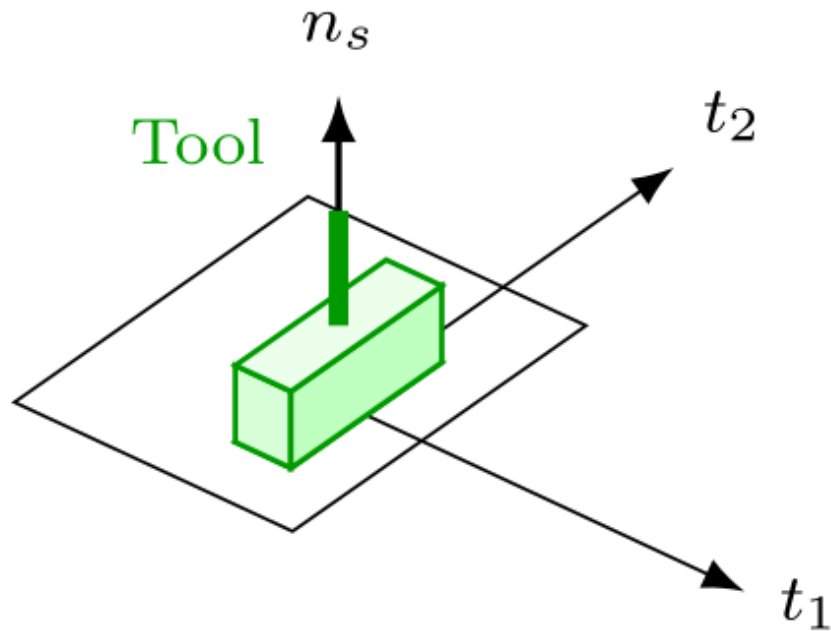
Orientation: 3 rotations



Forward kinematics: $x_{EE} = x_{EE}(q)$

Surface frame

- Surface axes define force and moment components and the directions for impedance gains.



n_s : upward surface normal

t_1, t_2 : perpendicular tangents

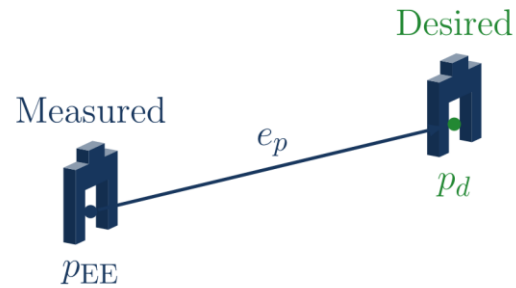
Cartesian impedance law

Force:

$$f = K_p e_p + D_p (\dot{p}_d - \dot{p}_{EE})$$

Positional error:

$$e_p = p_d - p_{EE}$$

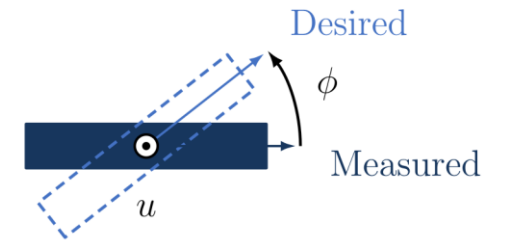


Moment:

$$m = K_R e_R + D_R (\omega_d - \omega_{EE})$$

Rotational error:

$$e_R = \phi u$$



Cartesian wrench:

$$F = \begin{bmatrix} f \\ m \end{bmatrix} = \underbrace{\begin{bmatrix} K_p & 0 \\ 0 & K_R \end{bmatrix}}_{K_c} \begin{bmatrix} e_p \\ e_R \end{bmatrix} + \underbrace{\begin{bmatrix} D_p & 0 \\ 0 & D_R \end{bmatrix}}_{D_c} \begin{bmatrix} \dot{p}_d - \dot{p}_{EE} \\ \omega_d - \omega_{EE} \end{bmatrix}$$

- In the surface frame, K and D are diagonal and decouple translation and rotation.

Real-time impedance control loop

Commanded joint torques (motors):

$$\tau = J^T(q)F$$

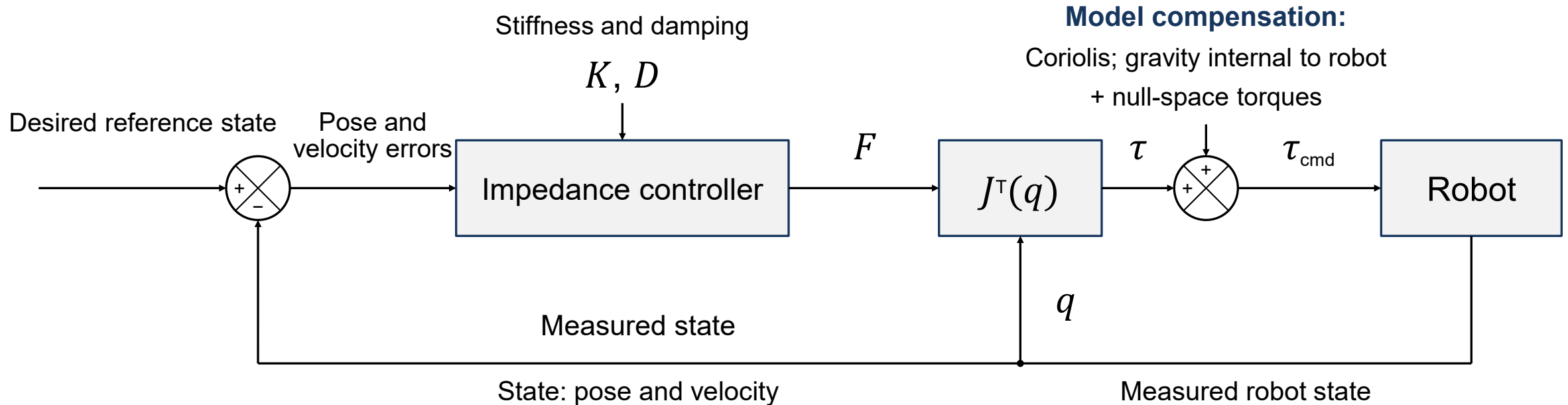
Cartesian contribution

Jacobian:

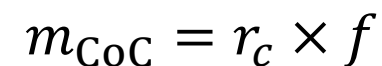
$$\dot{x}_{EE} = J(q)\dot{q}$$

Linear and angular velocity (geometric)

- Real-time control with a 1 ms cycle:

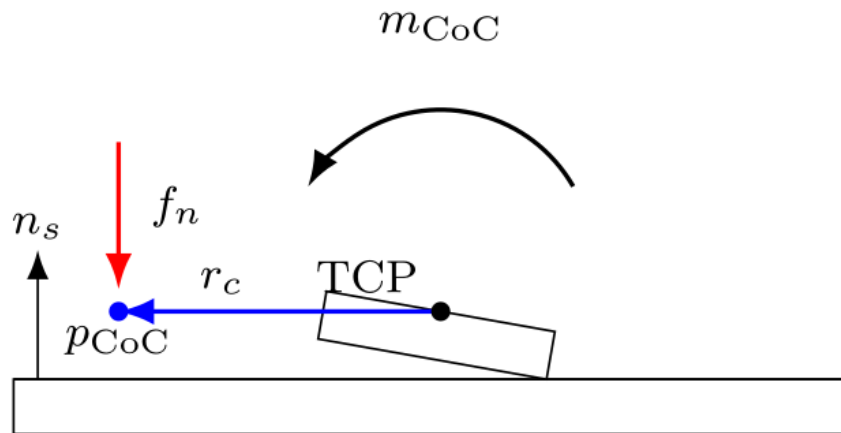


- The force acts as if applied at the virtual CoC.
- Additional moment about the TCP:

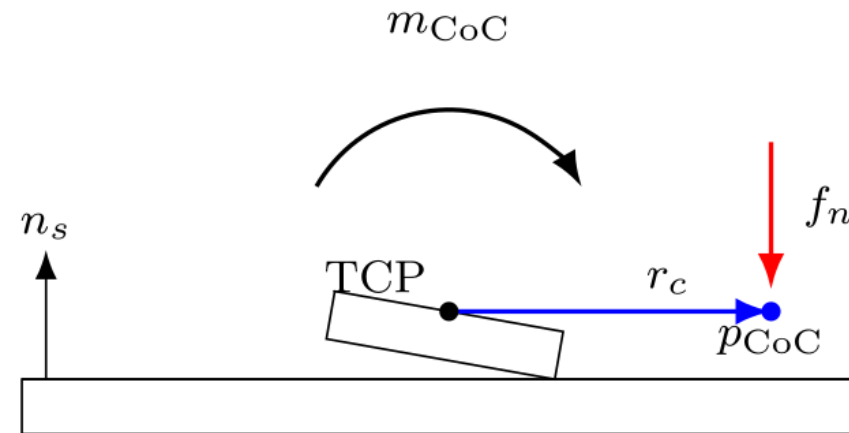


Supporting and opposing CoC moments

- Effect of CoC displacement:
- Added moment can support or oppose rotation.



(a) Supporting moment



(b) Opposing moment

Curved arrows show the moment about the TCP from the force at the virtual CoC.

Translation–rotation coupling

$$r_c = p_{\text{CoC}} - p_{\text{TCP}} \quad \text{Displacement from TCP to CoC}$$

$$A = \text{Ad}(r_c) \quad \text{Point-shift adjoint matrix}$$

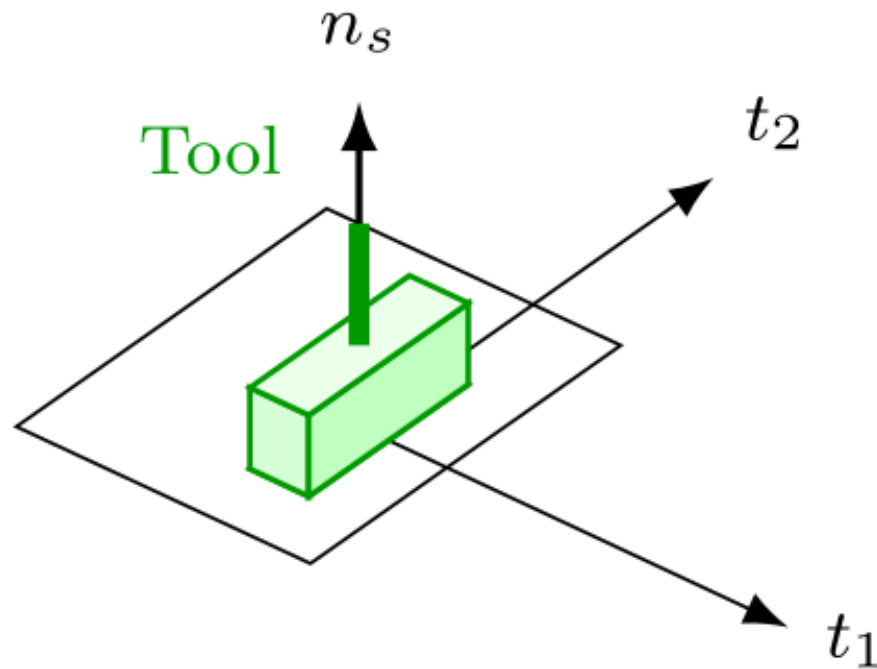
$$F_{\text{TCP}} = \begin{bmatrix} f \\ m \end{bmatrix} = \underbrace{A^\top \begin{bmatrix} K_p & 0 \\ 0 & K_R \end{bmatrix} A}_{K_{\text{TCP}}} \begin{bmatrix} e_p \\ e_R \end{bmatrix} + \underbrace{A^\top \begin{bmatrix} D_p & 0 \\ 0 & D_R \end{bmatrix} A}_{D_{\text{TCP}}} \begin{bmatrix} \dot{p}_d - \dot{p}_{\text{EE}} \\ \omega_d - \omega_{\text{EE}} \end{bmatrix}$$

- Non-zero off-diagonal blocks in K and D couple translation and rotation.

Contact phases and stiffness settings

- Experiment phases:

Tool orientation → Surface approach → Contact establishment → Surface grinding



- Compliant contact settings:

- Normal translation (compliant):

$$K_{p,n} = 350 \text{ N/m}$$

- Tangential translation (stiff):

$$K_{p,t_1} = K_{p,t_2} = 2000 \text{ N/m}$$

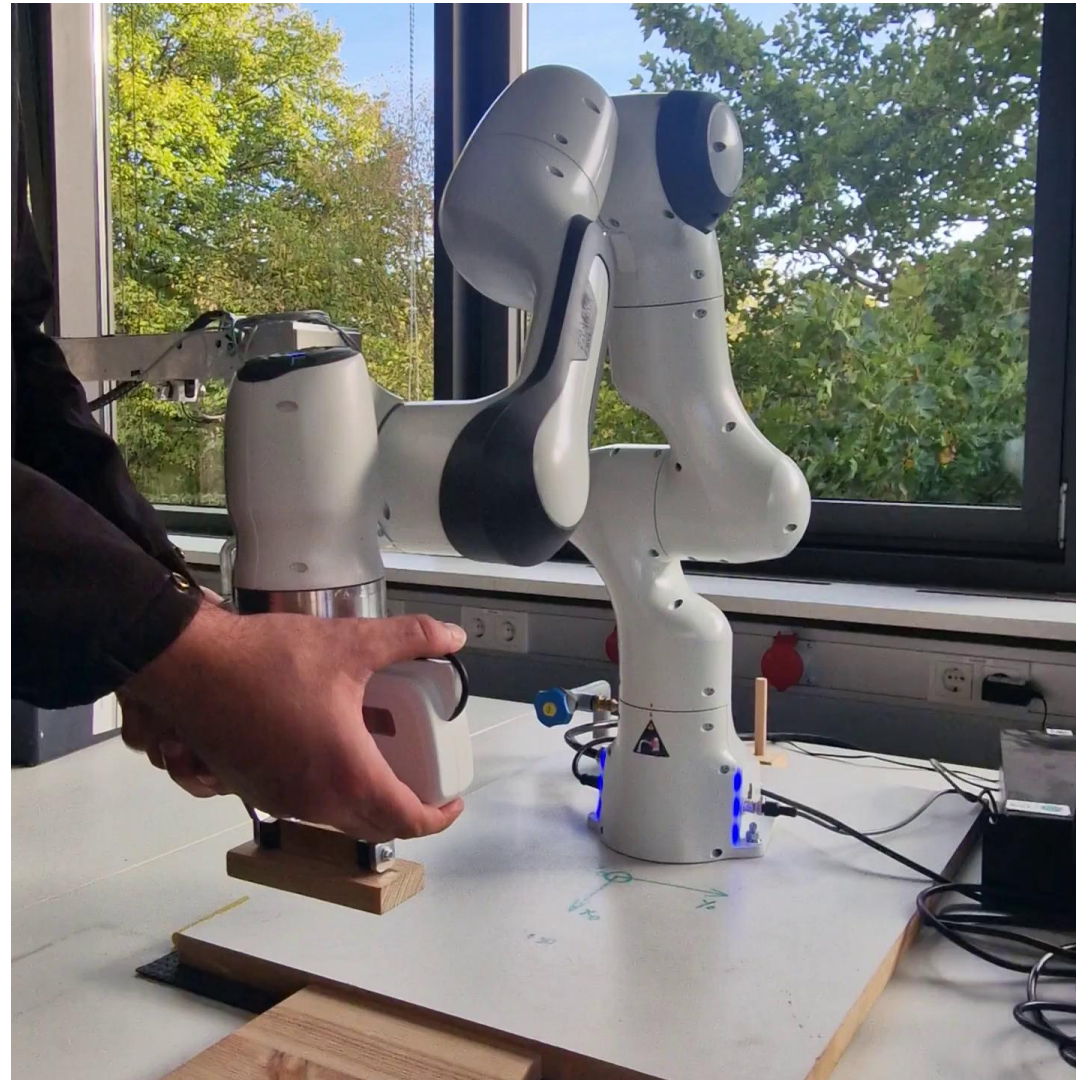
- Rotation about the surface normal (stiff):

$$K_{R,n} = 50 \text{ N m/rad}$$

- Rotations about the surface tangents (compliant):

$$K_{R,t_1} = K_{R,t_2} = 5 \text{ N m/rad}$$

Impedance response to manual displacement



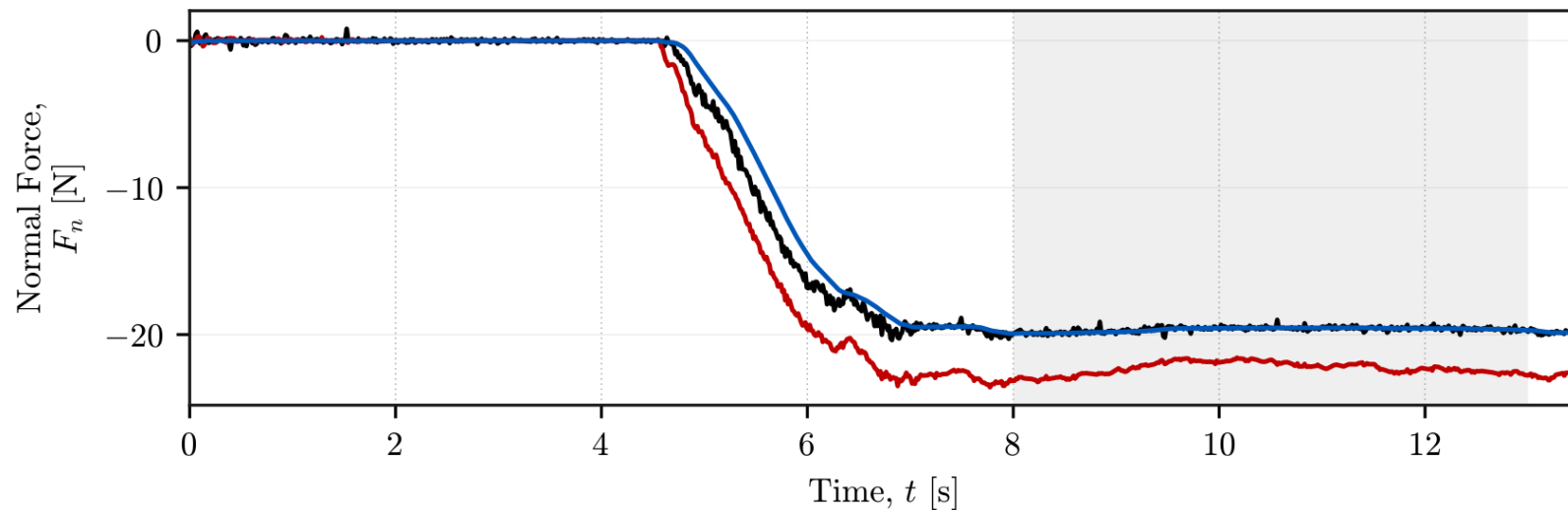
Normal-force plausibility assessment

Manual push: about 20 mm

Quasi-static calculation:

$$F_{n,qs} = K_{p,n} n_s^T (e_p - e_{p,0})$$

$$= (1000 \text{ N/m}) \cdot (-19.65 \cdot 10^{-3} \text{ m}) = -19.65 \text{ N}$$



— Commanded

— Model-estimated

— Quasi-static prediction

Quasi-static mean:

−19.65 N

Commanded mean:

−19.64 N

Model-estimated mean:

−22.29 N

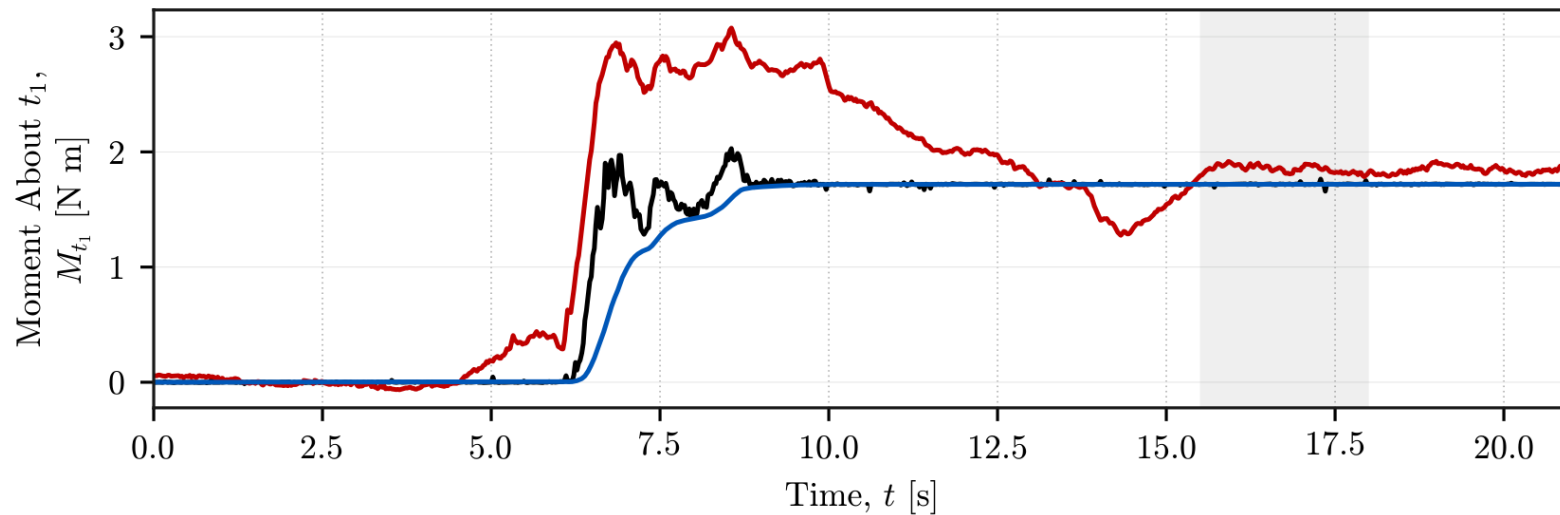
Moment plausibility assessment

Manual rotation: about 6.5°

Quasi-static calculation:

$$M_{t_1,qs} = K_{R,t_1} t_1^\top (e_R - e_{R,0})$$

$$= (15 \text{ N.m/rad}) \cdot \left(6.57 \cdot \frac{\pi}{180} \text{ rad} \right) \approx 1.72 \text{ N.m}$$



— Commanded

— Model-estimated

— Quasi-static prediction

Quasi-static mean:

1.72 N.m

Commanded mean:

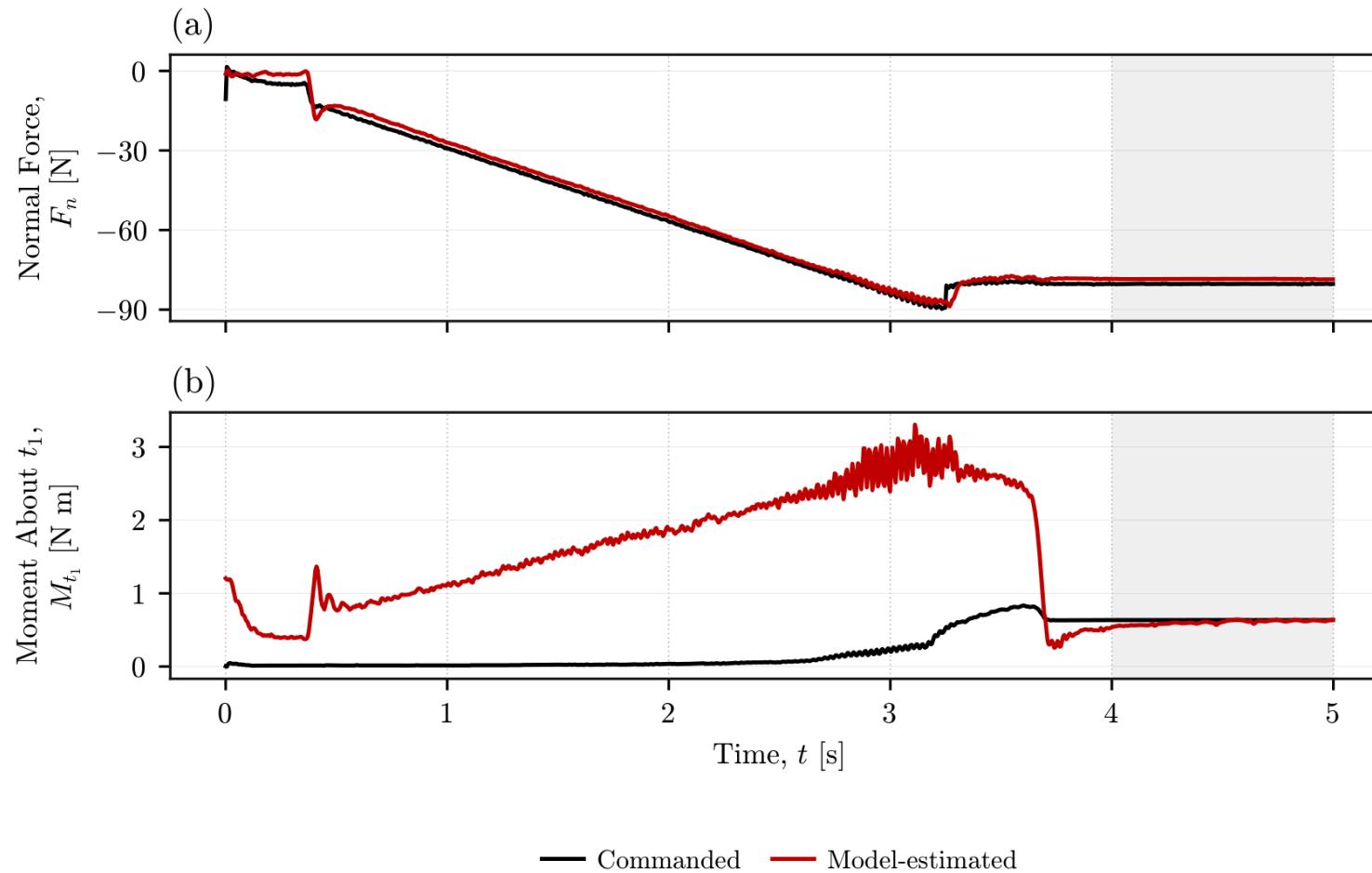
1.72 N.m

Model-estimated mean:

1.85 N.m

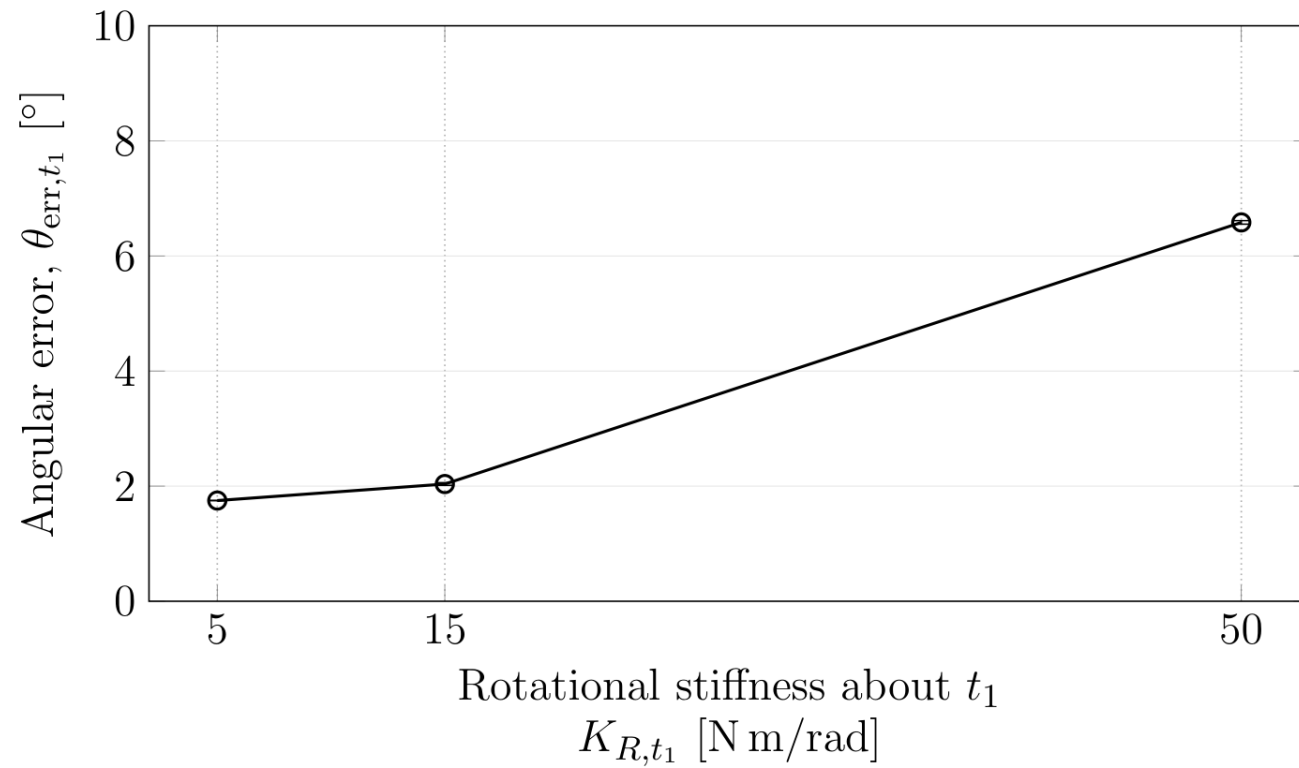
Force and moment during contact alignment

Commanded and model-estimated wrench:



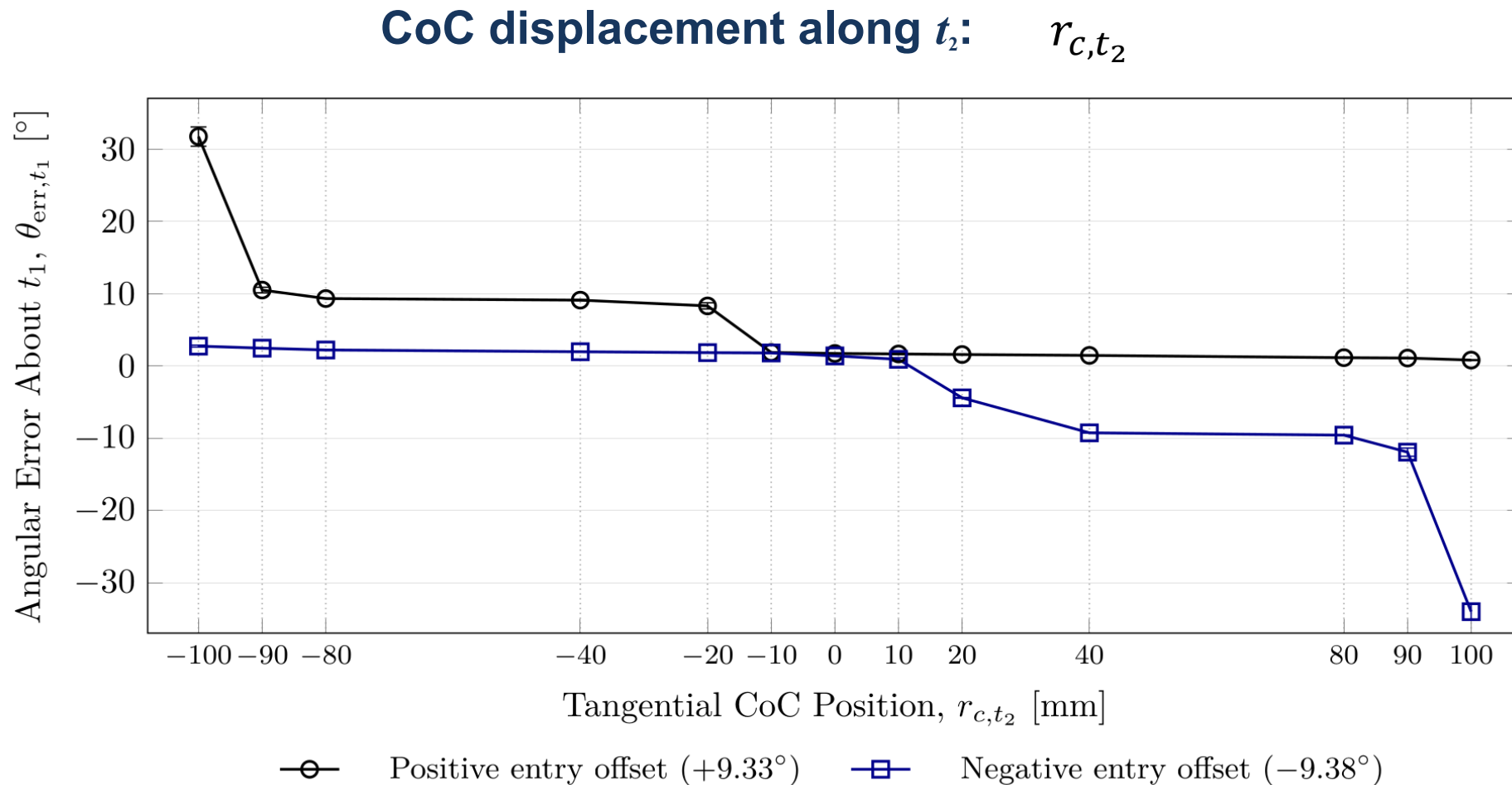
Rotational stiffness and angular error

Resistance to rotation about t_1 :



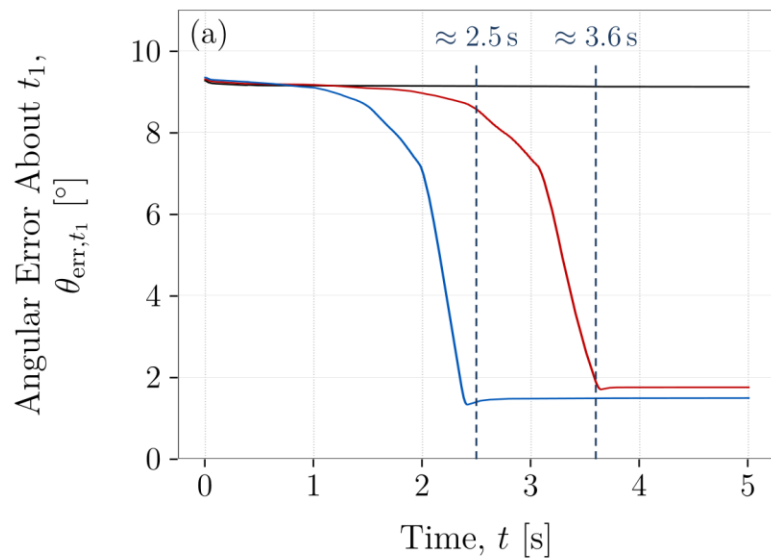
—○— Measured angular offset
 $\theta_{meas,t_1} = (9.31-9.32)^\circ$

Effect of CoC position on angular error

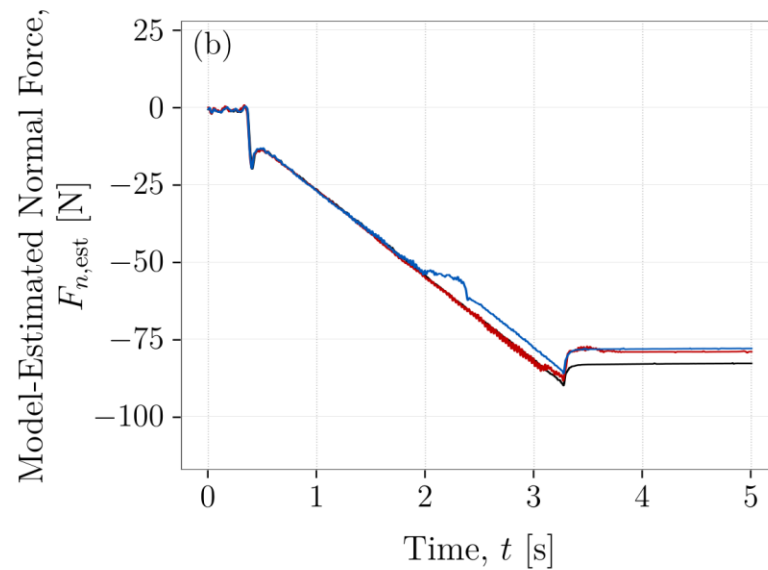


Contact response at different CoC positions

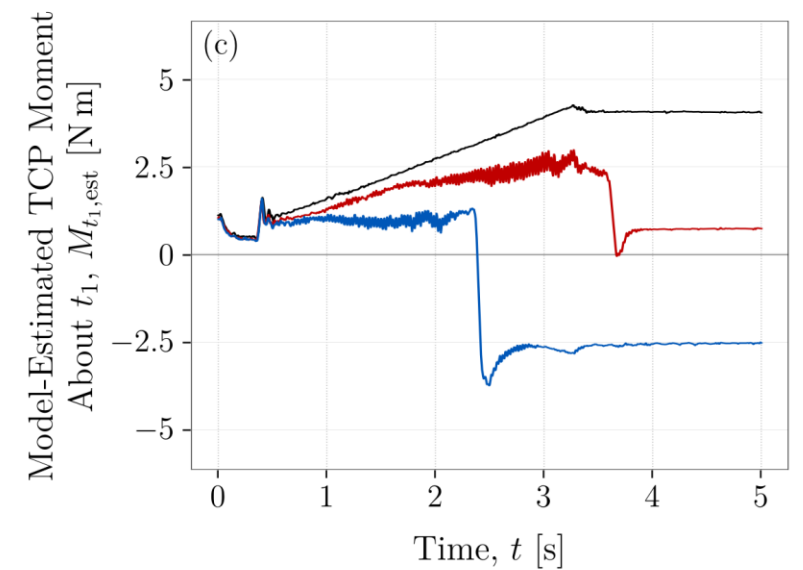
Angular error:



Normal force:



TCP moment:



— $r_{c,t_2} = -40$ mm
 — TCP: $r_{c,t_2} = 0$
 — $r_{c,t_2} = +40$ mm

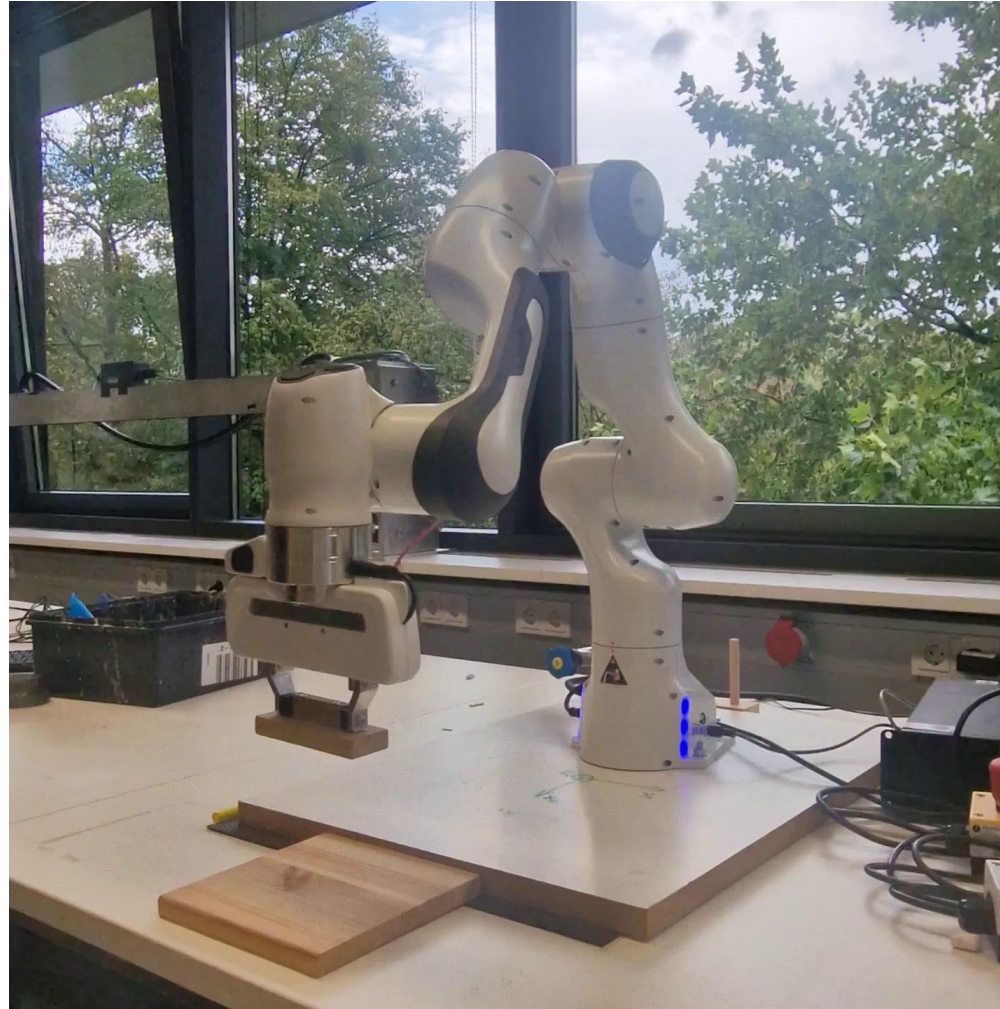
Null-space control: damping and conditioning

- Joint motion with a fixed tool pose:
 - The robot's extra degree of freedom allows its posture to change while keeping the (EE) pose unchanged.

Null space dimension = $7 - 6 = 1$ $\longrightarrow$ Null-space condition: $J(q) \dot{q}_{\text{null}} = 0$

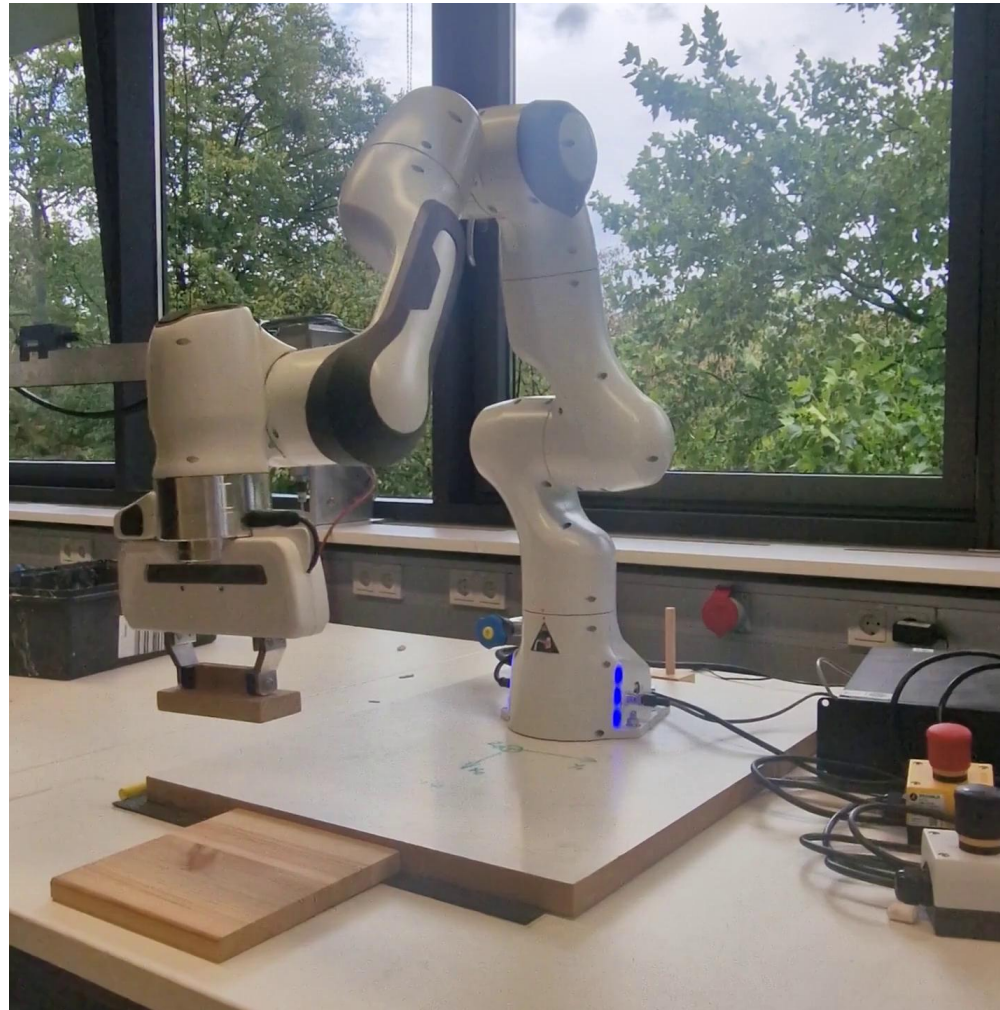
- Damping torque τ_d :
 - $\longrightarrow$ Opposes and damps null-space velocity.
- Conditioning torque τ_σ :
 - $\longrightarrow$ Adjusts the joint configuration through null-space motion to increase the minimum singular value $\sigma_{\min}$ of J and move away from singularities.

Null-space damping demonstration



Null-space control OFF → Damping ON

Null-space conditioning demonstration



Conditioning ON

Null-space experiment: setup and conditions

- Cartesian pose hold with commanded disturbance:

Hold the captured position and orientation.

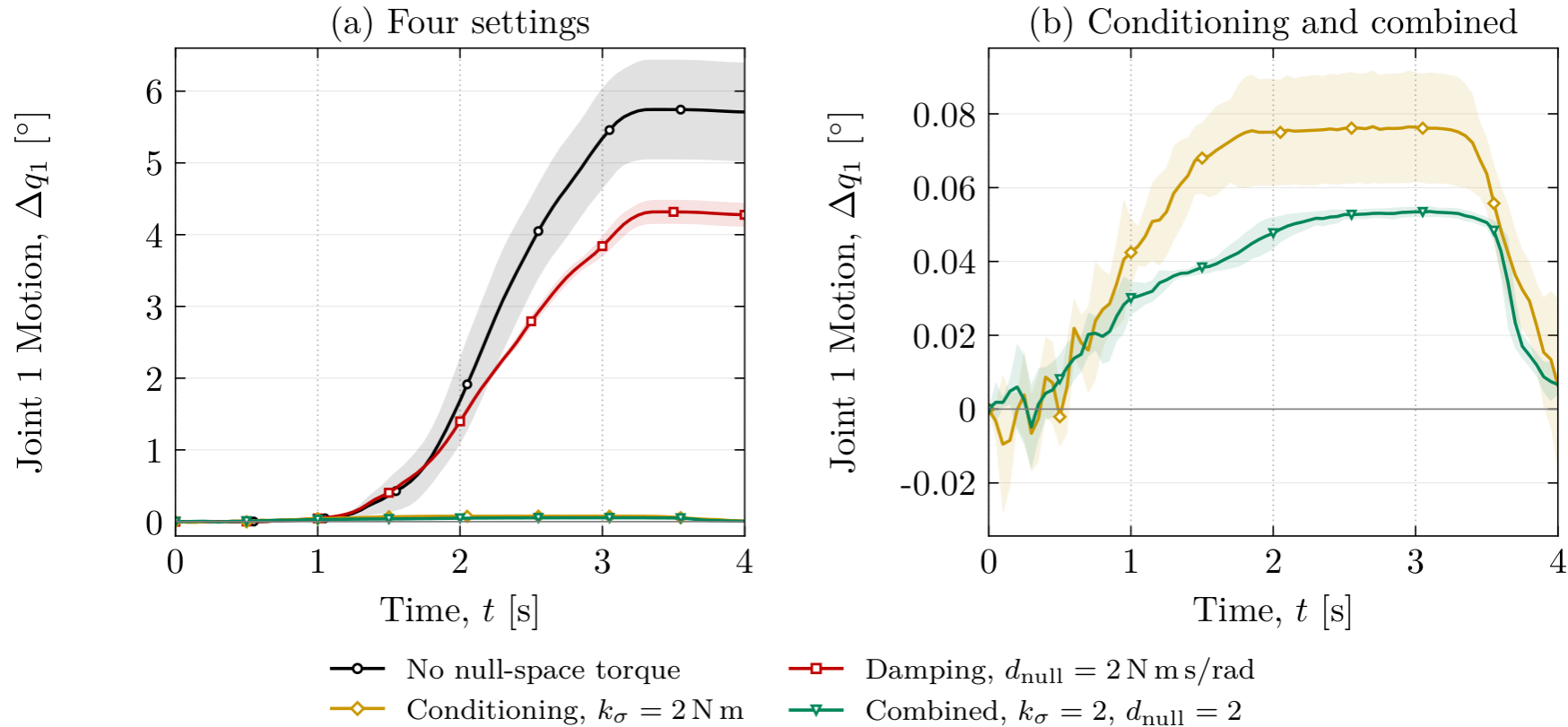
$$\tau_{\text{dist}} = J(q)^{\top} f_{\text{dist}}$$

Joint-torque equivalent of a virtual 20 N point force on link 3.

Joint 1 motion is shown because it receives the largest peak disturbance torque.

- Four conditions with three trials each:
 - Disturbance without null-space control torque
 - Conditioning: $k_{\sigma} = 2 \text{ N m}$
 - Damping: $d_{\text{null}} = 2 \text{ N m s/rad}$
 - Combined: conditioning + damping $k_{\sigma} = 2 \text{ N m}$ and $d_{\text{null}} = 2 \text{ N m s/rad}$

Joint 1 motion under null-space control

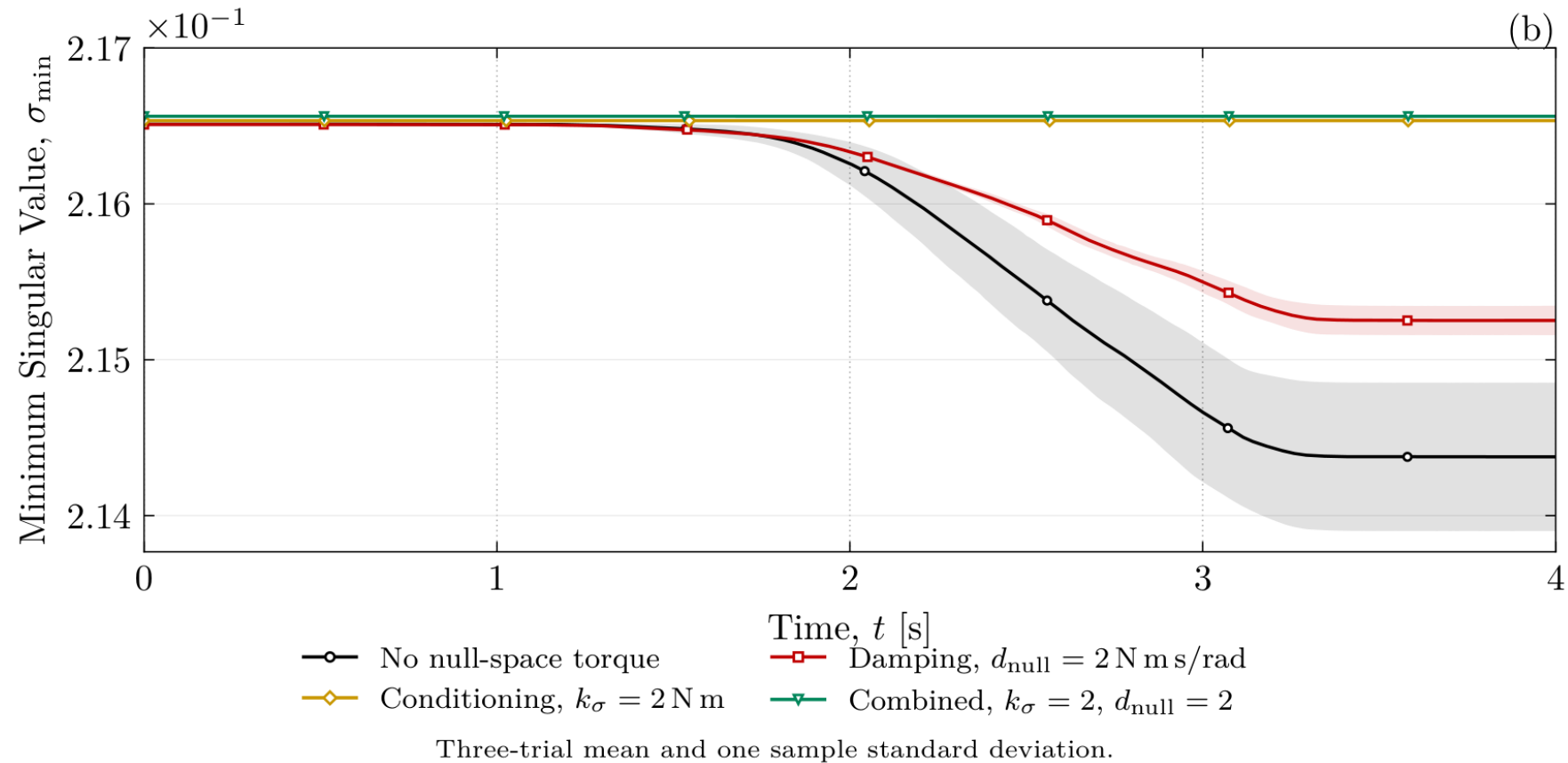


Three-trial mean and one sample standard deviation.

Conditioning (2 N m): repeated joint-motion reversals

Added damping: less total motion, remaining reversals

Jacobian conditioning: minimum singular value



Decreasing $\sigma_{\min}$: no null-space torque / damping alone

Nearly constant $\sigma_{\min}$: conditioning / combined control

Conclusion and future work

- Compliant rotation allowed contact moments to reduce angular error
- Higher rotational stiffness left a larger angular error
- The CoC could support or oppose the contact moment
- Null-space control reduced unwanted joint motion under disturbance
- Future work:
 - Use a more rigid tool mount for better measurements
 - Adapt the CoC to support the contact moment, then return it to the TCP after alignment

Opposing CoC moment during contact



Null-space torques and projector

Null-space projector:

$$N_\tau = I_7 - J^\top J^{+\top}, \quad \tau_{\text{null}} = \tau_d + \tau_\sigma$$

Projects secondary joint torques into the kinematic null space.

J : joint $\rightarrow$ (EE) velocities

J^+ : pseudoinverse

I_7 : 7×7 identity matrix

Damping torque:

$$\tau_d = -d_{\text{null}} N_\tau \dot{q}$$

Slows joint motion in the null space.
Zero at zero joint velocity.

d_{null} : damping coefficient [N m s/rad]

$\dot{q}$: measured joint velocity

Conditioning torque:

$$\tau_\sigma = k_\sigma N_\tau (s_\sigma v_7)$$

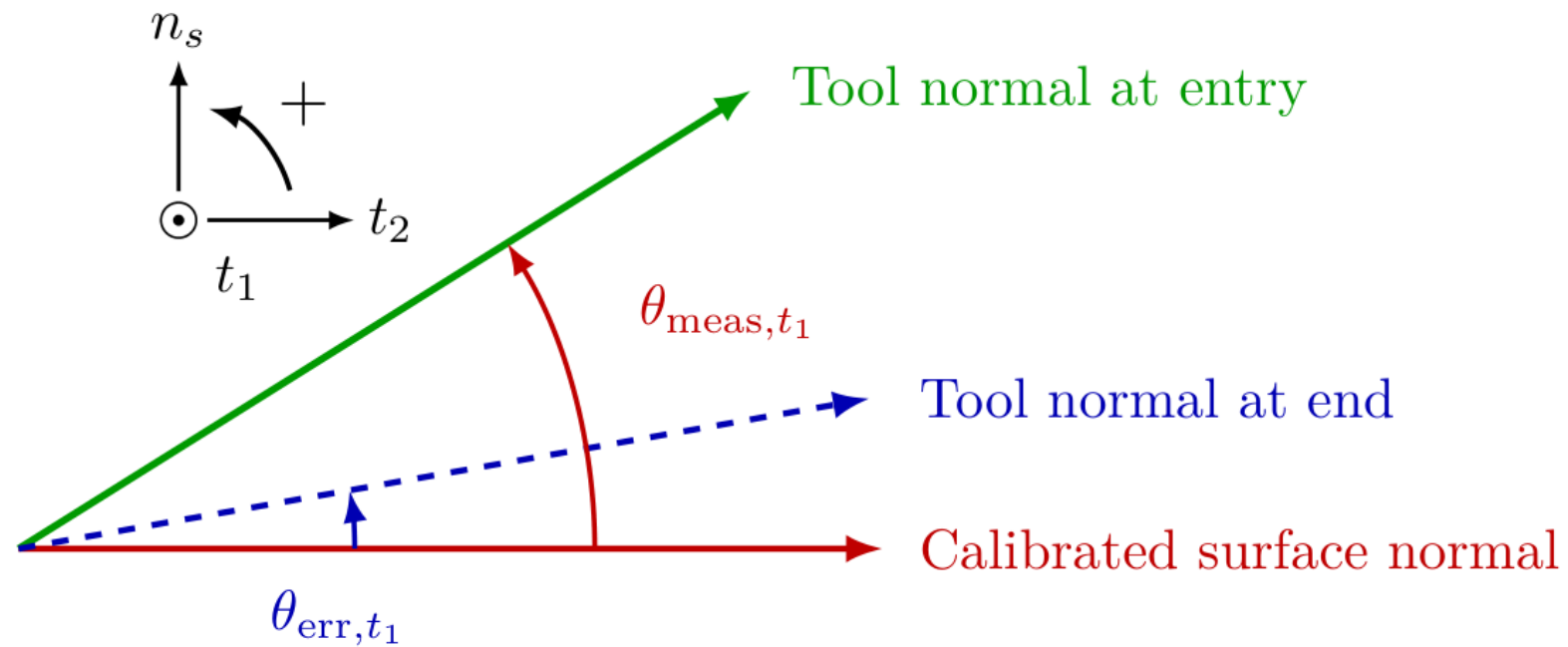
Seeks better local conditioning.
Can act at zero joint velocity.

k_σ : torque magnitude [N m]

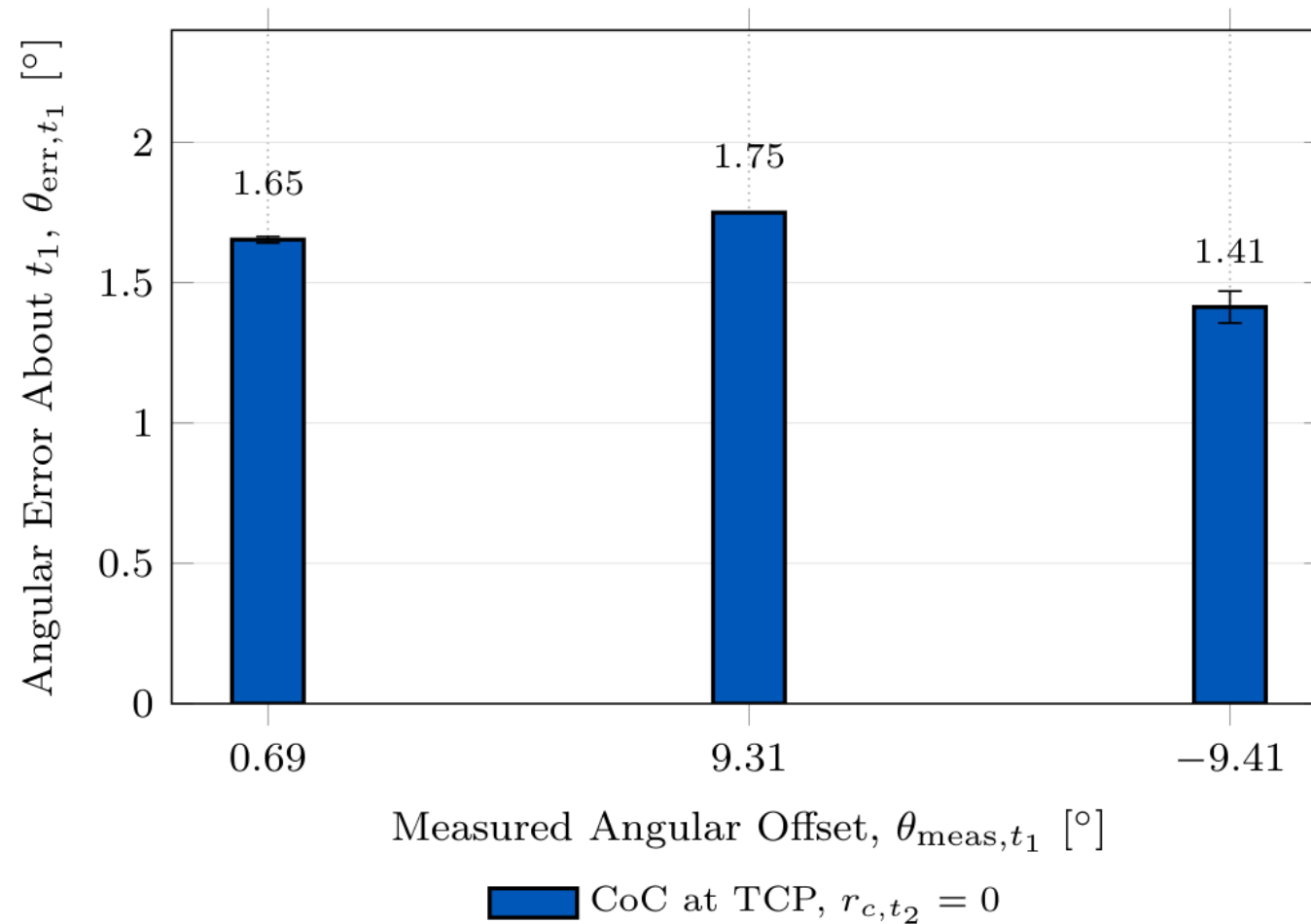
v_7 : unit null-space direction

s_σ : direction choice, zero in deadband

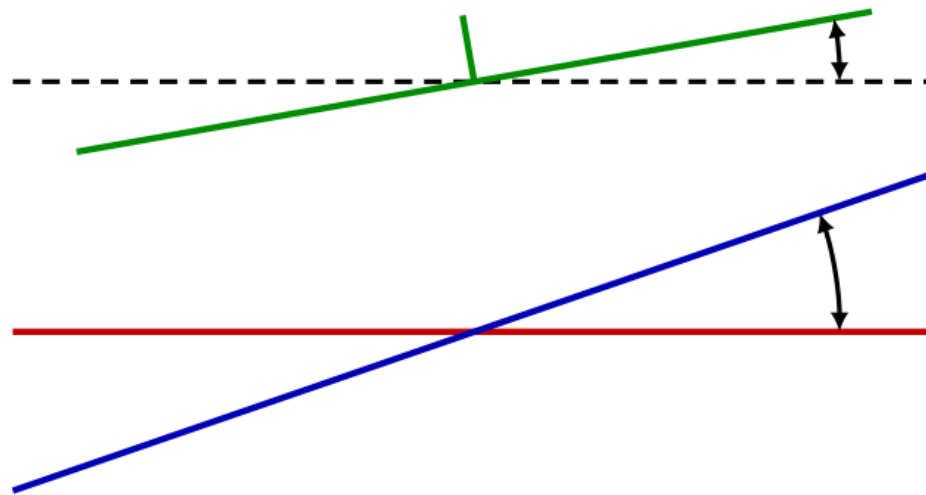
Entry offset and remaining angular error



Angular error with CoC at the TCP



Sources of angular offset



Desired–achieved difference

Configured–physical difference

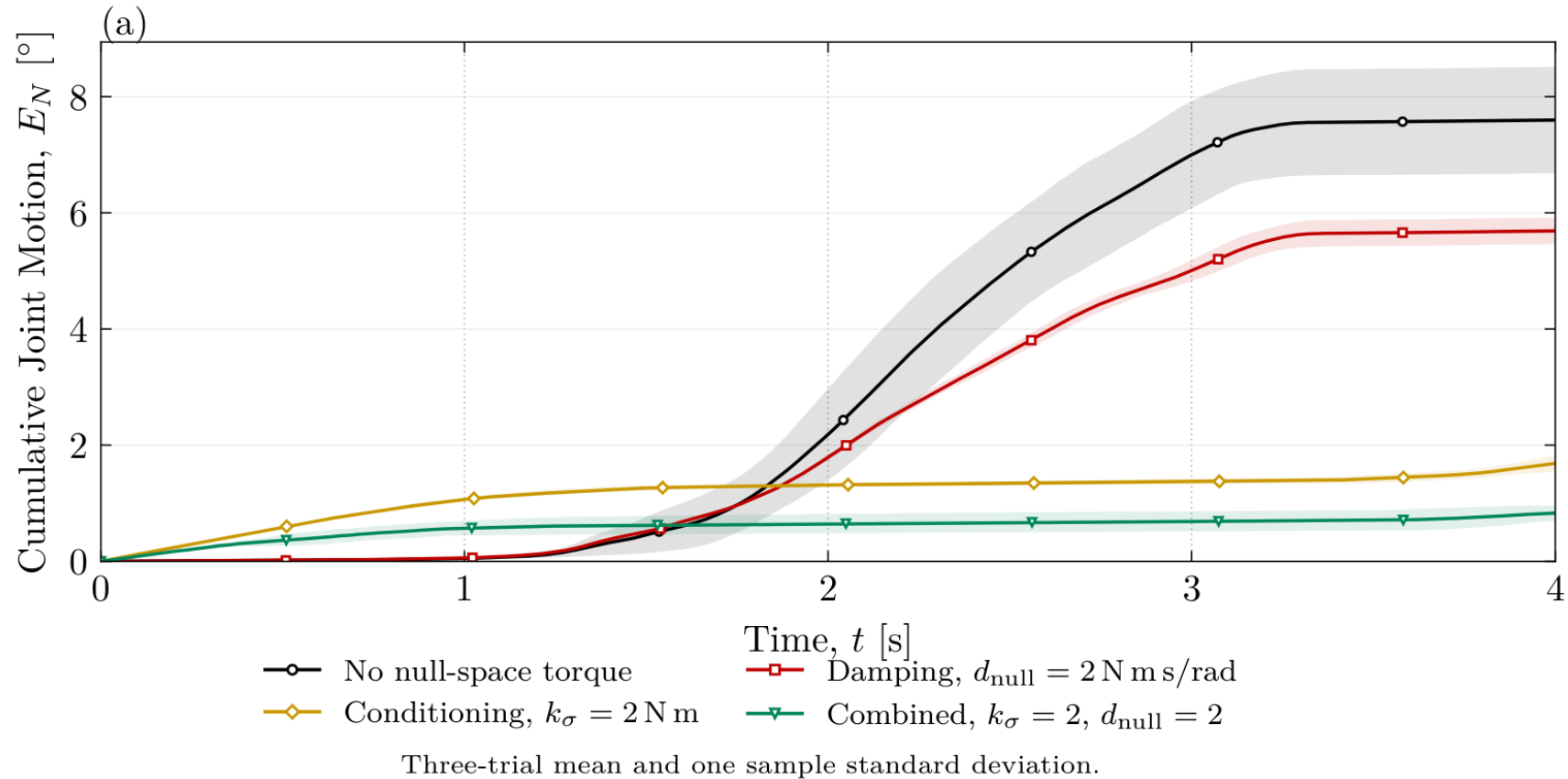
— Configured surface

- - - - Desired orientation

— Physical surface

— Tool face

Damping reduces cumulative joint motion



≈25% less cumulative motion with damping alone

≈50% less cumulative motion with damping added to conditioning